Fundamentals of Machine Learning
Understanding Mock (Lectures 1 to 5)

No derivations, no long computations. Situations you have not seen: what is going on, what is the issue, what would you do

Fundamentals of Machine Learning

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The pattern. Read the two tutorials again and count: more than half of the sub-questions are not calculations. They come in seven shapes, and every problem below uses them on a situation the tutorials never showed.

- **S1 Audit.** "A team did the following. Identify every mechanism that makes the result too optimistic, then propose a correct protocol." (T1 P4, T2 P8)
- **S2 Does it necessarily mean?** "Does this necessarily mean the labels are wrong?" (T1 1b)
- **S3 Distinguish.** "Why is this different from . . . ?" Shortcut versus leakage, cost threshold versus imbalance correction, prior shift versus costs, OOB versus out-of-fold, λ versus γ. (T1 1e, 2d, 6f; T2 5c, 6c, 8b)
- **S4 Diagnose from results.** A table or a curve; say what is wrong with each model and what to do. (T1 P5, T2 P7)
- **S5 Design an experiment that separates** two explanations of the same observation. (T2 9a)
- **S6 Trust and deployment.** "What results would make you trust it? Why would they still not establish . . . ?" (T2 9b–c, T1 4b)
- **S7 Available at decision time?** Leakage, future information, the right grouping key. (T1 1d, 8d–e)

The four-line answer that earns full marks on every shape

1. **Name the mechanism** in course words (shortcut, leakage, variance, prior shift, double counting. . .).
2. **Say what it does to the number**: which way the reported figure moves and why.
3. **Say what evidence would confirm it**: the check, the plot, the split you would run.
4. **Give the fix**, and if the question allows, its cost or its limit.

A one-word verdict ("overfitting") with no mechanism earns almost nothing. Two short sentences with the mechanism earn most of the marks.

Instructions. Two and a half hours, 100 points. Answer in sentences. Where a number helps, a one-line estimate is enough. State your assumptions.

1. The phishing detector that was too good [12 pts]

A company trains a phishing-email classifier. Phishing examples come from a public research corpus collected in 2015–2017. Legitimate examples are 2026 emails from the company's own servers. Features: the subject and body words, the `X-Mailer` and `Date` headers, the number of links, the HTML structure. With a random split, validation accuracy is 99.4%. In its first week of deployment the classifier flags almost nothing, while employees report several phishing emails a day.

(a) Give three things the model could have learned that separate the two sources but have nothing to do with phishing. [3 pts]

(b) Is this target leakage? Explain the difference precisely, using the example. [2 pts]

(c) Design one experiment, using only the data already collected plus a few hundred hand-labelled 2026 phishing emails, that tells whether the model detects phishing or detects the source. [3 pts]

(d) A colleague argues: "99.4% accuracy proves that the features we use are what makes an email phishing." Answer them. [2 pts]

(e) What should change in the data collection, not in the model? [2 pts]

2. A tree that cannot reach zero training error [12 pts]

A network team fits a fully grown decision tree (no depth limit, one observation per leaf allowed) to 1,000,000 labelled flows. The training error is 0.15, not 0. Validation error is 0.17. Each flow is described by 12 features: duration, packet and byte counts, ports, protocol, flags.

(a) A fully grown tree can isolate any single point. What must be true of the training data for its training error to stay at 0.15? [3 pts]

(b) Does this necessarily mean that 15% of the labels are wrong? Give a reason why it might, a reason why it might not, and a check on the data that would tell the two apart. [3 pts]

(c) The team proposes XGBoost with 5,000 rounds and depth 10. Predict what happens to training and to validation error, and explain. [2 pts]

(d) What kind of change can lower the 0.15, and why can no choice of model do so? Give one concrete example for network flows. [2 pts]

(e) Someone adds a column `flow_id` (a unique number per row, in the order flows were logged). Training error falls to 0. Explain why, and what happens to validation error under a random split and under a temporal split. [2 pts]

3. Four learning curves, four stories [12 pts]

Four teams report what happened while they increased the training-set size n, keeping the model fixed. For each, diagnose the situation and say what you would do next. A one-line answer per item is not enough: name the mechanism.

(a) Team A: training and validation error are both around 0.30, very close to each other, and almost flat from $n = 2{,}000$ to $n = 200{,}000$. [3 pts]

(b) Team B: training error 0.02, validation error 0.25 at $n = 5{,}000$; the validation error is still falling steeply at the largest n they have. [3 pts]

(c) Team C: validation set fixed (January traffic). They add February, then March traffic to the training set. Validation error *increases* each time. [3 pts]

(d) Team D: a strongly regularised logistic regression. Training error 0.12, validation error

0.09, with 400 validation flows. [3 pts]

4. Same model, new company, useless alerts [12 pts]

An intrusion detector is trained at company P, where 5% of flows are attacks. It is deployed unchanged at company Q. Q's analysts confirm that, on the attacks they know about, the detector catches the same fraction as at P. But precision falls from 0.8 to below 0.3, and the analysts are drowning in false alerts.

(a) Give a mechanism that makes precision collapse while recall is unchanged, with no change in what attacks or benign traffic look like. [3 pts]

(b) Does the model have to be retrained? Say what you would change instead, and under which assumption that is enough. [2 pts]

(c) Q has no labels yet. How could you tell, from the model's scores and inputs at Q alone, whether this is prior shift or whether attack or benign traffic itself looks different at Q? [3 pts]

(d) The manager proposes retraining on a 50/50 resampled dataset "so that the model takes attacks seriously". What does resampling do to the posterior the model outputs, and why is it a different operation from moving the threshold for costs? [2 pts]

(e) Q's team can investigate only 50 alerts a day. Which decision rule should now be used, and does the calibration of the scores still matter for it? [2 pts]

5. When probabilities lie [10 pts]

(a) A logistic regression trained on flows of at most 50 MB outputs $\hat{p} = 0.995$ for a 40 GB flow. Why does a discriminative model become *more* confident far from its data, and why is this confidence not evidence? [2 pts]

(b) How can a generative model of the same data signal that this flow is unusual? What would you do with such a flow operationally? [2 pts]

(c) Model A has accuracy 0.95 and log loss 0.30. Model B has accuracy 0.93 and log loss 0.18. The alert rule uses $C_{FP} = 2$, $C_{FN} = 8$. Which numbers matter for choosing, and which comparison would you actually run before deciding? [2 pts]

(d) A Naive Bayes model uses 20 features, ten of which are the same byte count expressed in different units or windows. Its posteriors are almost all 0.0001 or 0.9999. Explain, and say whether this harms accuracy, cost-sensitive decisions, both or neither. [2 pts]

(e) A student reads $\hat{p} = 0.7$ as "70% of flows like this one are attacks". When is that reading correct, and how do you check it? [2 pts]

6. Ensembles that do not behave as advertised [12 pts]

(a) Bagging 200 logistic regressions gives exactly the validation error of one logistic regression. Bagging 200 deep trees improves on one tree a lot. Explain the difference. [2 pts]

(b) A random forest has OOB error 0.06 but test error 0.15 on traffic from a later month. Each attack episode produced hundreds of near-identical flows. Give two separate reasons for the gap, and say which one a grouped validation set would remove. [3 pts]

(c) In gradient boosting, validation loss rises after round 200 while training loss keeps falling. In a random forest, adding trees from 200 to 2,000 never makes validation error rise. Explain why the number of trees is a capacity knob for one and not for the other, and what you

would do for the booster. [3 pts]

(d) The forest's MDI importance ranks `src_port` (the random ephemeral port the client's operating system picks) first. Why is that suspicious, why does MDI produce it, and what check would you run? [2 pts]

(e) In a stacking experiment the meta-model puts nearly all its weight on the deepest tree. What was probably done wrong, and how do you fix it? [2 pts]

7. Geometry problems [10 pts]

(a) To "give the model more information", a team appends 50 columns of random noise to its 8 flow features. kNN accuracy drops sharply; a random forest barely changes. Explain both, including why the forest still loses a little. [2 pts]

(b) The destination port is fed to kNN as a plain number. Give two pairs of ports for which the resulting distances are misleading, and propose a better representation. [2 pts]

(c) A 1-NN classifier has training error 0. Does this show that it overfits? [2 pts]

(d) With a random row split, validation picks $k = 1$. With a split grouped by attack episode, it picks $k = 25$, and the error is much higher. Explain both results. [2 pts]

(e) The scaler was fitted on all data, test set included. How much does this matter for kNN, and how much for a decision tree? [2 pts]

8. Reading coefficients and selections [10 pts]

(a) L1-regularised logistic regression sets the coefficient of `bytes_sent` to 0 and keeps `bytes_received`; the two have correlation 0.98. A report concludes that bytes sent are irrelevant to attacks. Correct it, and say what you would expect on another training sample. [2 pts]

(b) Unstandardised features: `duration` in milliseconds has coefficient 0.0004, `is_port_22` has coefficient 1.9. Is the port feature "4,750 times more important"? [2 pts]

(c) A team with 60 labelled hosts and 8,000 candidate features keeps the 20 features most correlated with the label (computed on all 60 hosts), then runs 5-fold cross-validation and gets 92% accuracy. Would you believe 92%? What would you expect if the labels were random coin flips? Fix the procedure. [2 pts]

(d) Early stopping with patience 1 stops a noisy validation curve at round 14; with patience 30 it stops at round 160 with lower validation loss. What is the role of patience, and what is the risk at each extreme? [2 pts]

(e) A team sees test error peak when the number of parameters is about n, then fall again for larger models. The manager concludes "the U-shaped curve is wrong, always use the largest model". What is right and wrong in that? [2 pts]

9. Compromised accounts and labels that arrive late [10 pts]

A detector flags compromised user accounts from login sessions. Labels come from security investigations, which close on average ten days after the incident. Training data: all sessions from 1 January to 31 March, randomly split into training and test sessions. Features include the user's number of logins *in the month of the session*, the login country, and a learned embedding of the user ID. The goal is to protect employees who join in April.

(a) Identify three ways the reported test performance is too optimistic for the stated goal. [3

pts]

(b) Sessions from the last days of March are almost all labelled benign. Why, and what does the model learn from them? How would you handle it? [2 pts]

(c) Propose a training, validation and test protocol for the stated goal. [2 pts]

(d) The per-session precision and recall are excellent, yet the security team says the detector is useless: alerts arrive after the attacker has already finished. What does the evaluation fail to measure, and what would you measure instead? [1 pts]

(e) After deployment, analysts investigate only the sessions the detector flags, and only investigated sessions can receive an "attack" label. The team plans to retrain every month on the new labels. What goes wrong over time, and what is a minimal fix? [2 pts]